

Concepts of Operating Systems And Linux Operating Systems

Trainer: Kiran Jaybhave



← End user

IDE | Chrome | VLC | a.exe | ... | Shell ⇐ Appln s/w

① ~~Interface~~ between
End user → H/W
or
Appln s/w → H/W

② Control program

③ Resource manager and Resource Allocator

④ Bootable device : SSD / HDD / PD / CD / DVD.

O.S. ⇒ Core O.S + Utility s/w + Appln s/w

Core O.S → Kernel
- Windows (ntoskernel.exe)
- Linux (Vmlinuz)

Utility s/w ↳ Control panel / Disk mgr...

Appln s/w ↳ paint / edge / Notepad

CPU | RAM | HDD/SSD | Keyboard | monitor ⇐ Hardware Res^r

Keyboard: primary I/P device

Function of Core OS

1. Process mgmt
2. CPU scheduling
3. memory mgmt
4. File & I/o mgmt
5. HW abstraction

} Core

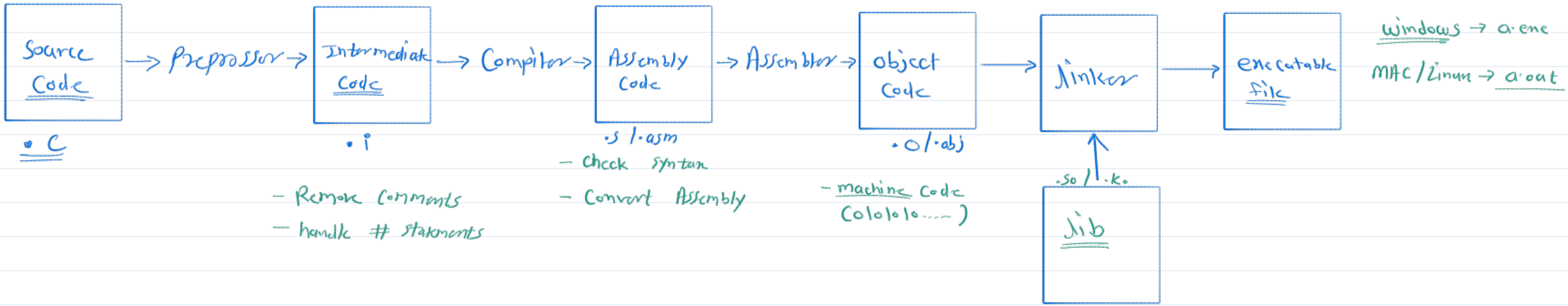
6. User interfacing
7. Networking
8. Protection & security.

} optional

* Compilation steps.

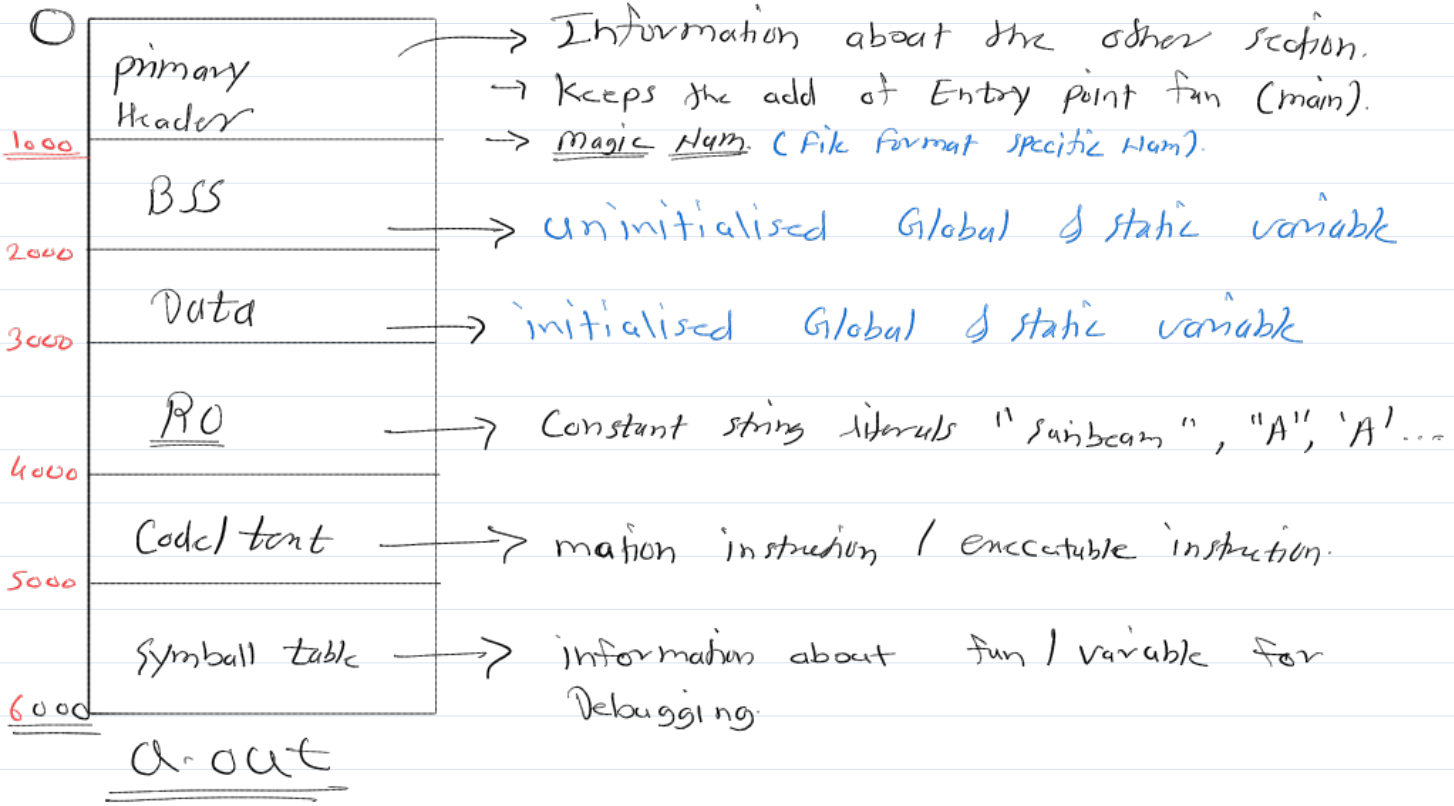
- .c / .cpp / .java => Source Code
  - ↓ Compile (gcc)
  - ↓ a.exe => executable file

Toolchain : gcc
- set of tools which work on source code one by one to convert it into machine code.
- Preprocessor
- Compiler
- Assembler
- Linker
- Debugger
- utilities



* Program Structre.

a.out => Sectioned Binary



→ Information about the other section.
→ Keeps the add of Entry point fun (main).
→ Magic Num. (File format specific Num).

→ uninitialised Global & static variable

→ initialised Global & static variable

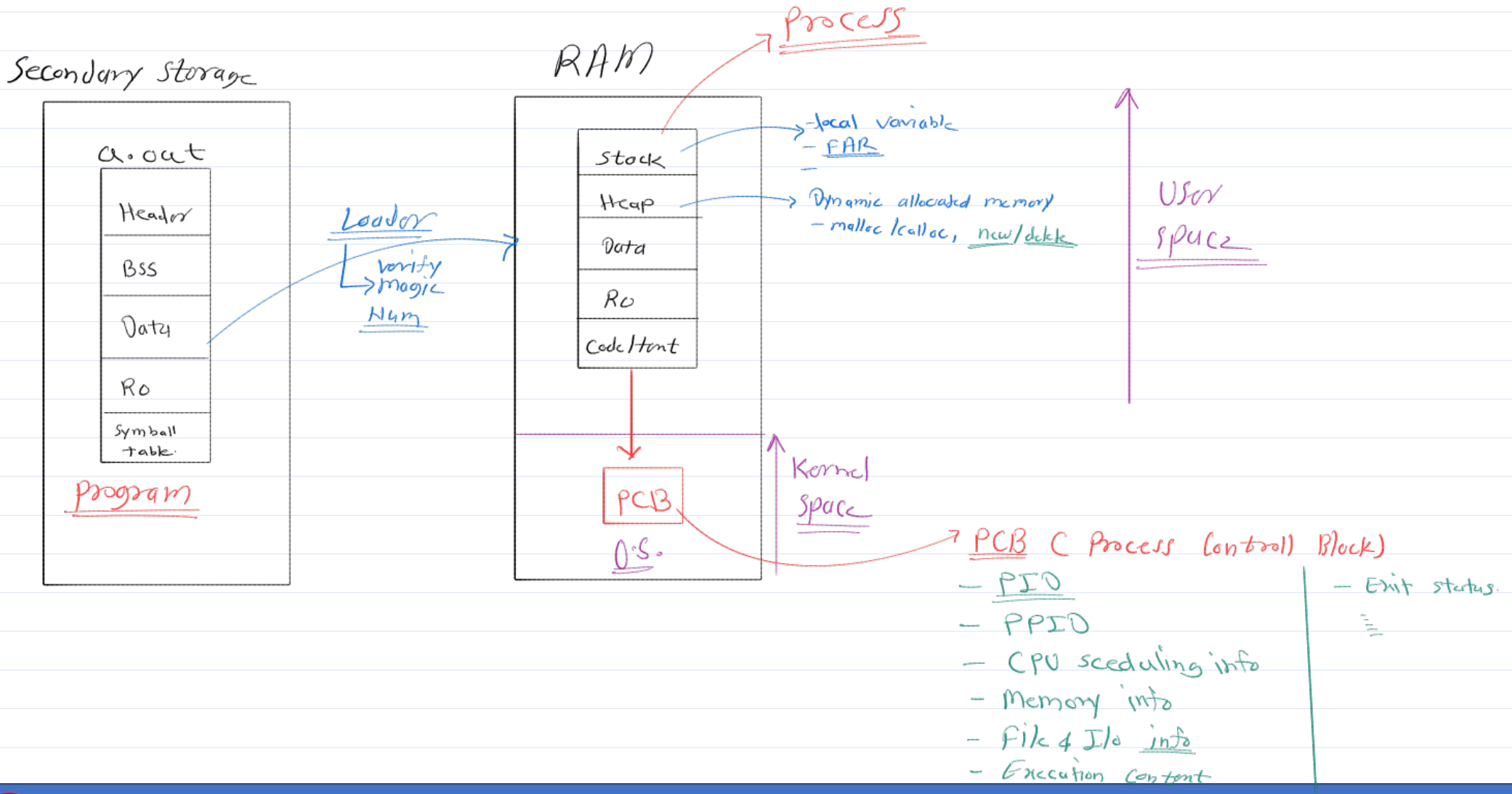
→ Constant string literals "Sunbeam", "A", 'A' ...

→ mation instruction / executable instruction.

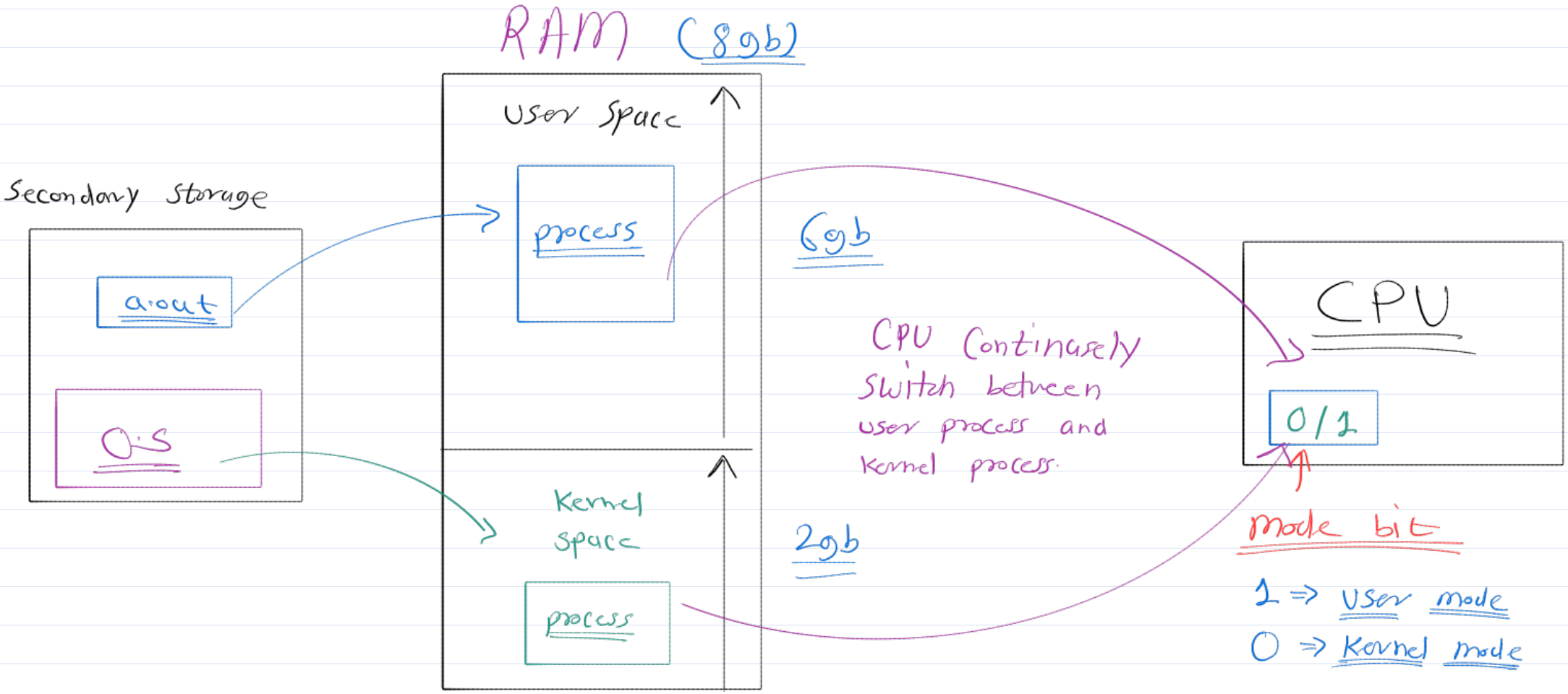
→ information about fun / variable for Debugging.

- .out → Linux (? ELF)
- .exe → windows (MZ)

Process


Secondary storage
a.out
Header
BSS
Data
Ro
Symbol table
Program
Loader
verify magic Num
RAM
Process
Stack
Heap
Data
Ro
Code/Text
local variable
- FAR
Dynamic allocated memory
- malloc/calloc, new/delete
User space
Kernel space
PCB
O.S.
PCB (Process Control Block)
- PID
- PPID
- CPU sceduling info
- Memory info
- File & I/o info
- Execution Context
- Exit status.

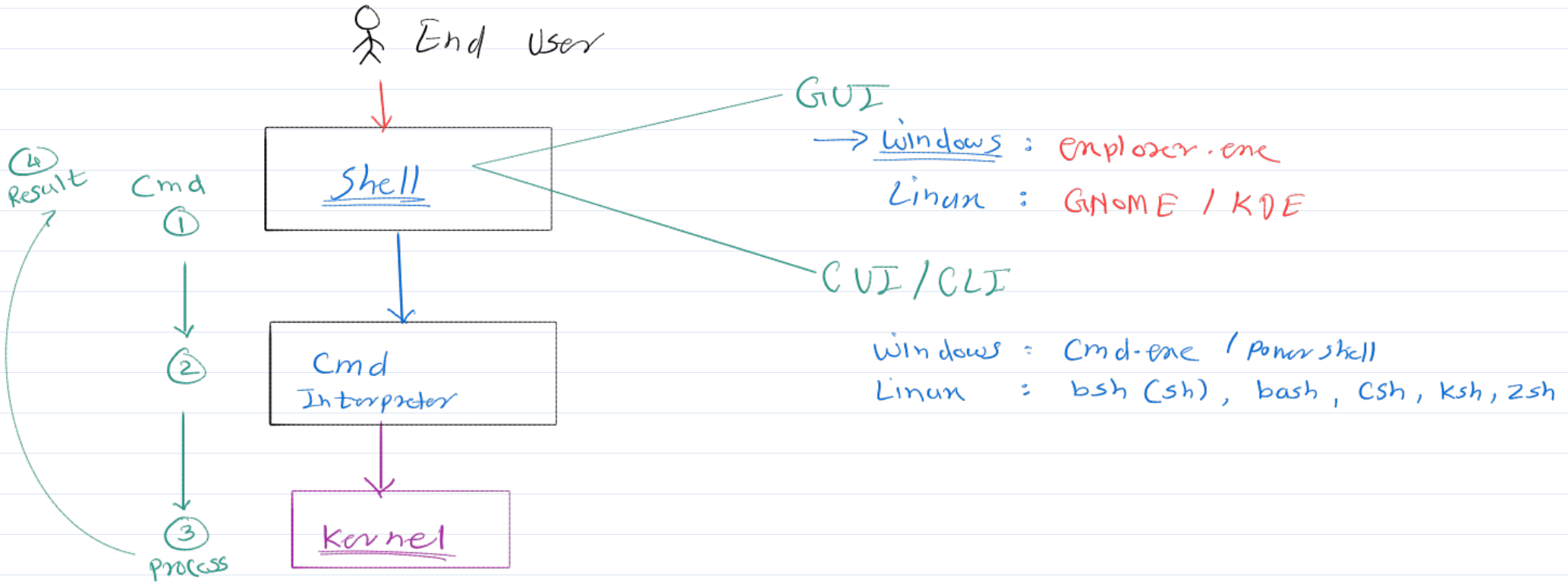
* Dual mode operation.



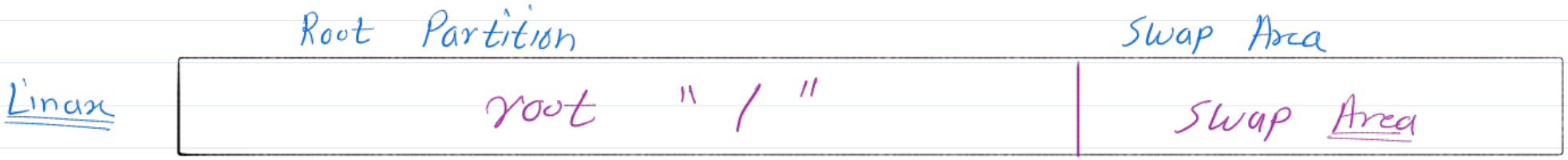
* User Interfacing

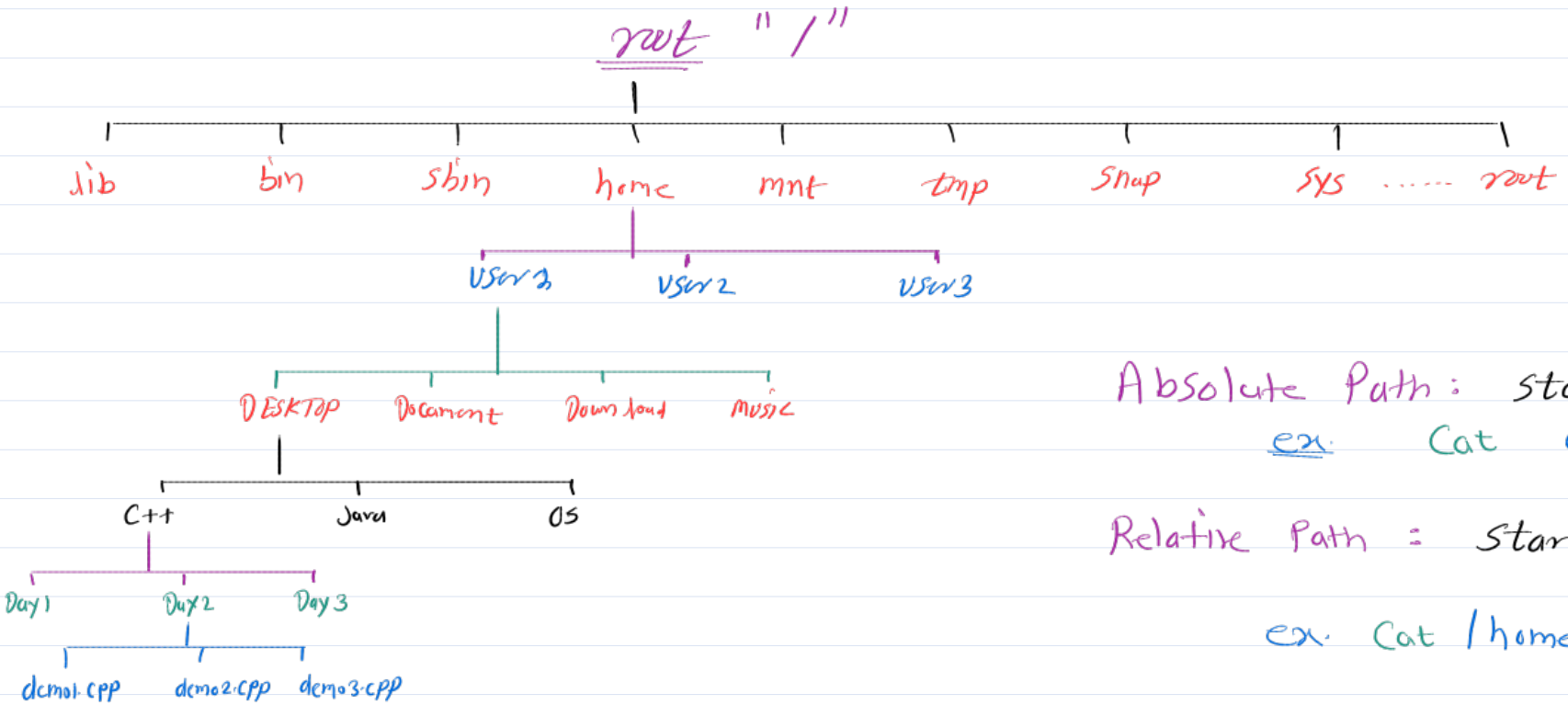
- User interfacing is provided by one of the program & it is called as shell.

- shell is an intermediate betn OS & user
- Shell is command Interpreter.



* Root File System





Absolute Path : start always from '/'.

ex. Cat C++ / Day-2 / Demo3.cpp

Relative Path : start from Current Working dir

ex. Cat /home/user1/Desktop/C++/Day-2/Demo3.cpp

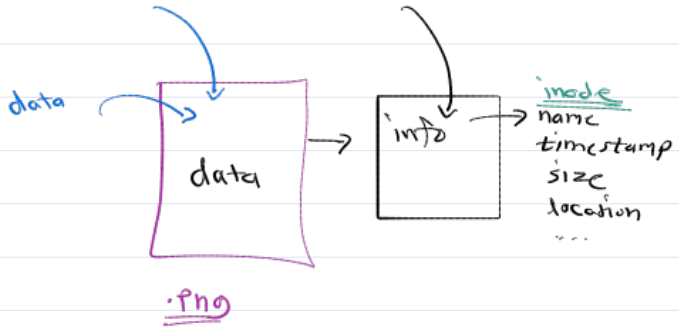
* Storage and Partition

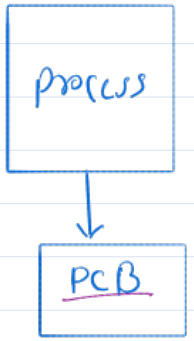
	Storage devices	Partition
Windows :	Disk 0 (SSD) Disk 1 (PD) Disk 2 (HDD)	C: , D: , E: ....
Linux :	sda (SSD) sdb (PD) sdc (HDD)	sda1, sda2, sda3 sdb1 sdc1, sda2

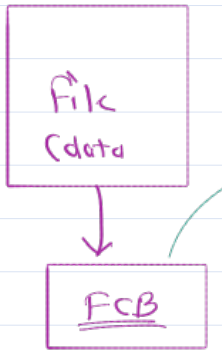
* File system.

File : collection of logically Related data.

System : File => data + metadata (Actual data) (Info about data)







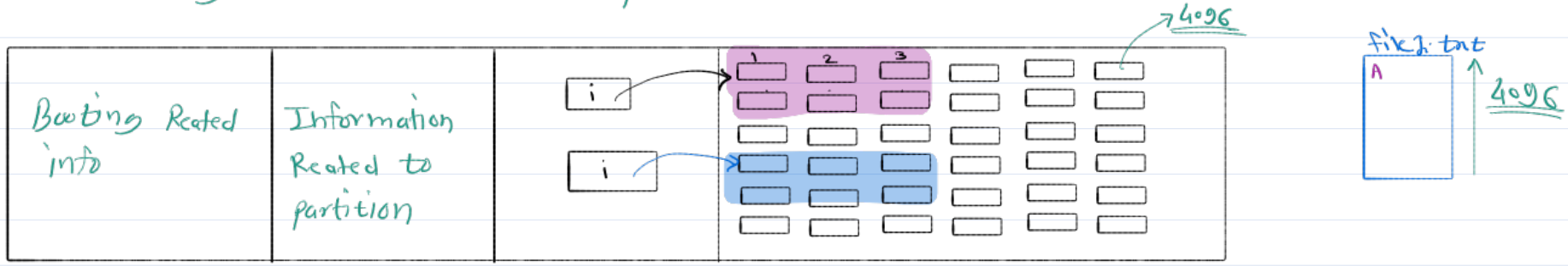
FCB / Inode

File Controll block / inode
- inode no
- File Name
- size
- timestamp
- Permission (r-w-x)
- Block

Check info in inode cmd : > stat file.txt

File System

- File system = organize files on partition



Boot block or boot sector
- → Boot loader Program
- → Boot strap Program

Super block or Volume control block
- Info about volume
  - Capacity
  - label
  - emply, used block cnt

inode table or master file Table → inode list.

Data block → Actual data.